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A 3-D gravity model for a volcanic crater in Terceira Island (Azores)

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SUMMARY

The seismic response of the São Sebastião volcanic crater, located at the SE end of Terceira Island (Azores), is characterized by an amplification of ground movements inside the crater with respect to the surrounding area. This variability of ground motion is also observed inside the volcanic depression itself. To gain insight into this phenomenon, a gravimetric survey with dense coverage of the region where the depression is located was carried out. Data corrections, including the calculation of terrain density by the Nettleton method and filtering of non-correlated anomalies (noise) by least-squares prediction, led to a Bouguer anomaly map with a standard deviation of 0.213 mGal. The analysis of this map and its interpretation is carried out by means of a stabilized non-linear inversion methodology resulting in a 3-D model of density contrasts, which identifies the geometry of the causative sources in the area. The gravity inversion outlines a correlation between the presence of scoria cones eruptive centres and the zones of negative density contrast in the calculated model. At São Sebastião village, another important structure with negative density contrast is located in places where a more destructive and strong anomalous behaviour has been observed during several seismic events. Furthermore, the gravity inversion seems to confirm a collapse model for São Sebastião crater and to indicate the presence of a deep conduit system compatible with the fissural nature of the volcanism in this area.

Key words: calderas, crustal structure, gravity anomalies, inverse problem, seismicity, volcanic structure.

INTRODUCTION

The Azores Archipelago is located at the triple junction of the North American, Eurasian and African plates. The first two are separated by the Mid-Atlantic Ridge (MAR), located between Flores and Faial islands, while the Eurasian and African plates are separated by a complex tectonic structure, the Azores–Gibraltar fault, which in the area of the archipelago comprises a 'microplate' (e.g. Luis *et al.* 1994), with a roughly triangular shape (Fig. 1). In general terms, the northern edge of this microplate includes a WNW–ESE spreading centre, the so-called Terceira Rift, incorporating Graciosa, Terceira and São Miguel islands and a series of *en echelon* rifted basins from the Formigas Trough at 36.8°N–24.5°W, to a point near 39.3°N–28.8°W. The archipelago shows recent volcanism and seismic activity located mostly along a series of volcanic ridges and major tectonic lineaments with a general WNW–ESE trend (Searle 1980), like the Faial-Pico Fracture Zone (FPFZ) and the S. Jorge fault (SJF) (Fig. 1).

Terceira Island, together with Graciosa, São Jorge, Pico and Faial islands form the Central Group of the Archipelago. Located in the Terceira Rift, this island shows active volcanism (with a wide variety of rocks compositions and structures) and important seismic activity.

From east to west, Terceira Island includes four central polygenetic volcanoes with caldera (Serra do Cume, Guilherme Moniz, Pico Alto and Santa Bárbara) and a basaltic fissural zone, mainly in the central and SE parts of the island (Fig. 2) (Nunes 2000). Volcanic products extruded from these central volcanoes, mostly in Quaternary times include ignimbrites, pumice fall deposits, strongly evolved lava flows (comendite to pantellerite lavas), scoria and spatter deposits, surtseyan tuffs and basaltic lava flows such as the 2115 years BP eruption (Fig. 2). After the settlement of the island (during the 15th century) three basaltic volcanic eruptions took place: the first one in 1761, the second in 1867 and the last one that started in late 1998 and continued until mid-2000. The 1761 eruption occurred on-land, east of Santa Bárbara Volcano, while the later two were located on sea, NW of the island at about 2.5 and 8.5 km offshore, respectively (Fig. 2).

Regarding seismicity, Terceira Island has been affected by several damaging earthquakes, such as the Praia da Vitória earthquakes in 1614 and in 1841, and the 1980 January 1 earthquake (Fig. 1), the strongest in Azores during the last century, with $M = 7.2$ (Hirn *et al.* 1980). The majority of these earthquakes and others (such as the 1997 and 1998 seismic crisis) were felt with an anomalous Mercalli modified intensity (MMI) distribution throughout the island, with

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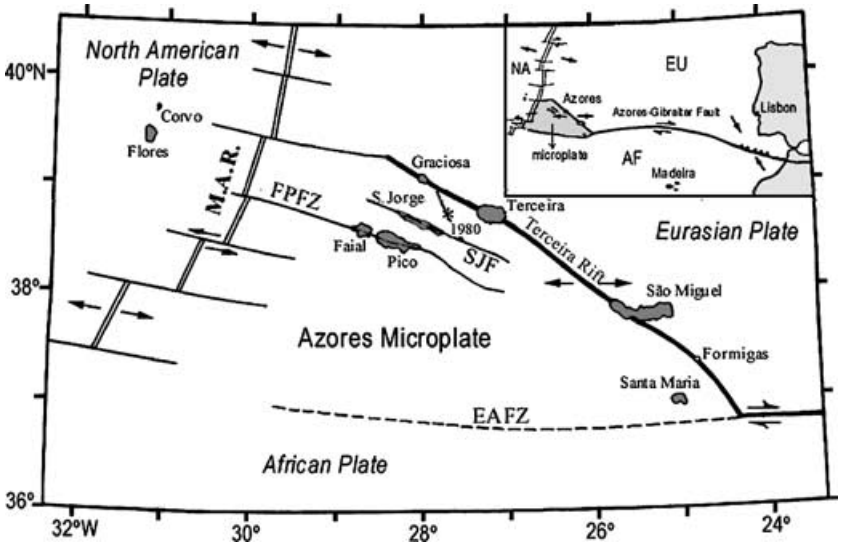

Figure 1. General geotectonic framework of Azores archipelago (modified from Nunes 1999). A star indicates the location of the 1980 January 1 earthquake epicentre, the strongest in the Azores during the last century, MAR = Mid-Atlantic Ridge, EAFZ = East Azores fracture zone (corresponding to the southwestern sector of Azores Gibraltar fault), FPFZ = Faial-Pico fracture zone, SJF = S. Jorge fault, EU = Eurasian Plate, AF = African Plate, NA = North American Plate.

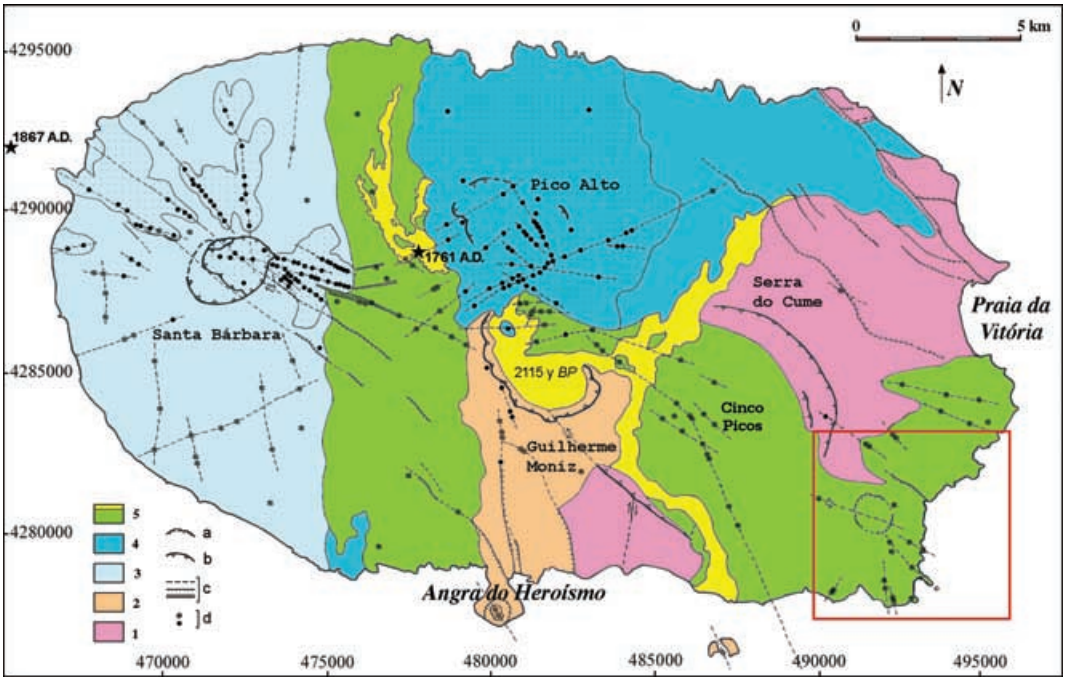

Figure 2. Volcano–tectonic sketch of Terceira Island: (1) Cinco Picos-Serra do Cume Volcano; (2) Guilherme Moniz Volcano; (3) Santa Bárbara Volcano; (4) Pico Alto Volcano; (5) basaltic fissural zone, including the 1761 AD and a 2115 years BP lava flow, in yellow. Dotted areas include domes and trachyte s.l. lava flows. (a) crater rim; (b) caldera rim; (c) volcanic and tectonic lineaments; (d) eruptive centres (simplified from Nunes 2000). Stars point out the eruptive centres of the 1761 AD and the 1867 AD volcanic eruption, the later about 2.5 km offshore of Terceira. SE inset square points outside the São Sebastião studied area. UTM coordinates are in metres.

higher intensity inside São Sebastião crater than in the neighbouring communities (Fig. 3) (Forjaz 1980a; Oliveira 1992; Oliveira *et al.* 1997, 1998).

Inside this volcanic crater, located in the SE of Terceira, with an average diameter of 1100 m and a depth of about 50 m, the village of São Sebastião has also shown an anomalous seismic behaviour in a more local expression, because some areas of the village have usually suffered more damage than others.

To obtain a better understanding of this anomalous seismic response at São Sebastião, we have carried out a structural study of the subsurface by means of gravimetric modelling, looking for a correlation among the geological, tectonic and volcanic characteristics of the zone. The lack of *a priori* information of the zone from other geophysical studies such as for instance seismic inversion, magnetism or measurements of rock densities in boreholes, poses some additional difficulties in defining the model for

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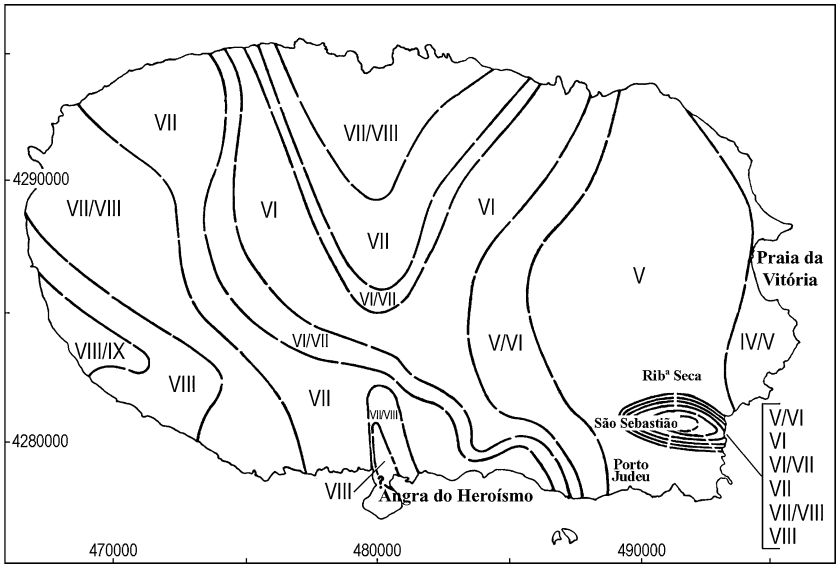

Figure 3. Isosseismal map of Terceira, 1980 January 1 earthquake, showing intensity anomalies (Mercalli modified) in São Sebastião village (simplified from Forjaz 1980a).

the area, but nevertheless the gravity results obtained seem very promising.

It is known that gravity data can offer valuable information concerning lateral changes in mass distribution for crustal–upper-mantle models (e.g. Dal Moro *et al.* 1998). Then, from this information it is possible to look for correlations between the gravity model and, for instance, a special structural instability (e.g. Hildenbrand *et al.* 1996), or tectonic and volcanic processes (e.g. Rousset *et al.* 1989; Camacho *et al.* 1997; Malengrau *et al.* 1999; Montesinos *et al.* 1999). However, it is not usual to find information concerning correlations between gravity data and local seismic amplification (a site effect).

The inverse gravimetric problem has, however, as its main disadvantage an intrinsic non-uniqueness in its solution (e.g. Al-Chalabi 1971). To reduce this ambiguity it is necessary to use *a priori* information (geological or geophysical), including mathematical stabilizing constraints that are obtained from the geological setting (e.g. Silva *et al.* 2001). So, we use prescribed density contrasts from geological data in a 3-D gravity inversion approach (Camacho *et al.* 2000, 2002) to determine the geometry of sources of the gravity anomaly field observed. In this case, the solution is obtained by the l_2-fitness to the observed gravity data and by two constraints on the source compactness and the density contrast values fixed for these sources (positive and negative values are accepted simultaneously). It is thus expected that these constraints produce geologically meaningful results, compatible with the information available for the area.

The use of this inversion technique for São Sebastião region allowed better knowledge concerning the subsurface structures present in the area and provided a clearer understanding of its volcanic and tectonic processes. In order to establish a model of the deeper structures present in the area of São Sebastião and also to obtain the subsurface local structure of the volcanic crater in more detail, this study was planned in two scales. A wider survey covering all the region surrounding the depression, until the south coast of the island, led to a regional structural model of the zone, whereas a denser distribution of the gravity stations inside the volcanic depression allowed us to study the shallower structures present in the village area.

GEOLOGY OF THE SÃO SEBASTIÃO AREA

A detailed geological study was carried of the São Sebastião village and neighbouring areas allowing a better understanding of its eruptive history and volcanological constraints (Nunes 2000), namely through the establishment of the detailed volcanostratigraphy of the area, the nature of its volcanism, the eruptive styles and the mapping of the major tectonic lineaments. The geological map obtained (Fig. 4) shows that, with the exception of the trachyte formations related to the Cinco Picos–Serra do Cume old volcano, all the mapped geological formations have a generally basaltic composition. This is the case for the recent Fonte Bastardo and Pico do Refugo basaltic lava flows, the former having a maximum expected age of 2000–3000 yr BP (Lloyd & Collis 1981), based on its morphological and geological features and the local stratigraphy.

The eastern and SE sectors of the area are dominated by extensive scoria deposits (Fig. 4), either as *lapilli*-layered deposits or as scoria cones of different sizes and ages (see Table 1). Among these, Cruzes and Pico dos Comos represent some of the oldest scoria cones in the area. These cones were responsible for the extrusion of very porphyritic basaltic lavas for which the high degree of weathering seems to be compatible with K/Ar ages of 3.5 Myr reported by Ferreira & Azevedo (1995) for lavas sampled NW of Baía da Salga. All of these old lavas are extensively buried by younger lava flows of 'intermediate age' (Fig. 4).

It is not possible to give a precise time interval for the extrusion of these 'intermediate-age' lavas, due to the lack of isotope age determinations in the area and the absence of charcoal remains underneath the mapped lava flows. However, during the mapping work it was possible to infer some stratigraphical relationships among the geological formations shown in Fig. 4 (Nunes 2000). We concluded, for example, that Pico das Contendas (a major strombolian-type eruption scoria cone) is one of the youngest 'intermediate-age' geological formations in the area, the eruption of which took place probably a few tens of thousands of years ago.

The southern and SE rims of the São Sebastião crater are cut on a *lahar* deposit and the depression is filled in its upper levels

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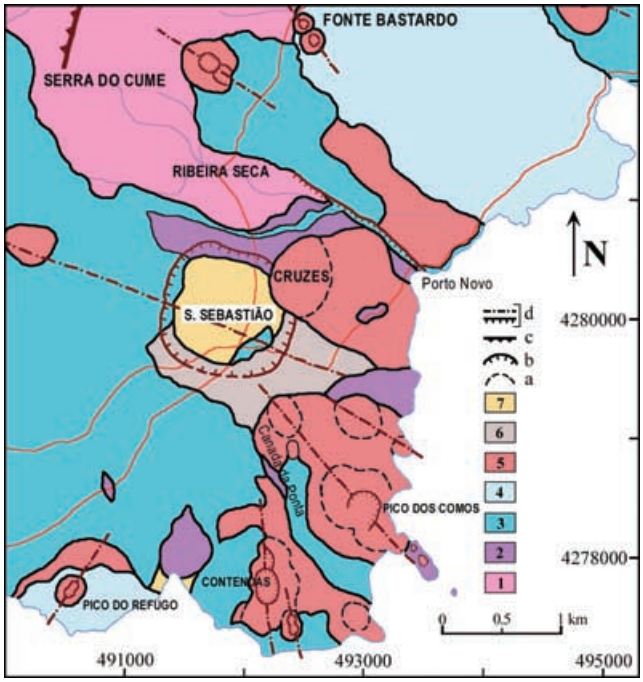

Figure 4. Volcanological map of the São Sebastião area: (1) old trachyte lava flows (Serra do Cume volcano); (2) old basaltic lava flows; (3) intermediate-age basaltic lavas; (4) recent basaltic lava flows; (5) basaltic scoria and *lapilli* deposits; (6) *lahar*; (7) fluvial and slope deposits; (a) scoria cone base; (b) crater rim; (c) caldera rim; (d) volcanic and tectonic lineaments (simplified from Nunes 2000). UTM coordinates are in metres.

Table 1. Correlation between geological, morphological and gravimetric evidence for several tectonic and volcanic structures at the studied area. (See Fig. 11 for reference locations).

Structure	Ref.	Geology	Morphology	Gravimetry
Pico Refugo	1	x	x	
Contendas 1	2	x	x	x
Contendas 2	3	x	x	x
Pta. Mina	4	x		x
Pico Comos	5	x	x	x
E Arrabalde	6	x	x	x
Arrabalde	7	x	x	x
'144' cone	8	x	x	x
'173' cone	9		x	x
Cruzes	10	x	x	x
Porto Novo fault	a	x	x	x
Canada da Ponta lineament	b			x
Pico dos Comos lineament	c	x	x	x
São Sebastião crater	d	x	x	x

mostly by slope deposits of different nature and composition, and by fluvial deposits. These products, together with pumice (from the nearby central volcanoes), scoria, ashes and other loose and detritical material, probably have a total thickness of about 38 m, as pointed out by a 182 m deep borehole located inside the depression, on its SE sector (Lloyd & Collis 1981).

The northern border of the São Sebastião depression consists of a porphyritic basaltic lava extending on land inside the Cinco Picos caldera, and seems to be the same as that found down to a depth of 38 m in the above-mentioned borehole. Additionally, no explosive deposit, either outcropping or buried, was found that could establish a hydromagmatic genesis for that depression, as suggested by Lloyd & Collis (1981). All of these observations and also the fissural and basaltic nature of the volcanism in this area of Terceira Island, suggest that the São Sebastião volcanic depression might be considered as a pit crater structure formed after a general collapse to about 80 m. The crater was probably formed during the Pleistocene as can be inferred from the ^{14}C radiometric date of 44 950 ± 1340 yr BP (Nunes 2000) obtained in charcoal remains found at depths of about 12 m in a borehole located inside the São Sebastião crater (Malheiro 1998).

The tectonic structures of the mapped area are defined mostly by volcano–tectonic lineaments, with different trends and in general terms radiating from the São Sebastião crater (Figs 2 and 4). Usually these lineaments are observed on the aerial photos of the area as alignments of volcanic cones, craters, vents and/or eruptive fissures, without a clear indication of its sense of motion. This is the case, among others, of the Pico das Contendas, Pico dos Comos and Pico do Refugo lineaments, inferred namely from stretched and/or double craters emplaced on these scoria cones. However, the most significant superficial tectonic feature in the area is the Porto Novo fault (Forjaz 1980b), with a dominant NW–SE trend and with the SW block down dropped to about 20 m. A probable WNW–ESE fault runs through the São Sebastião crater, with a left lateral component, as could be deduced from the crater rim displacement (Fig. 4). It seems that these last two structures mostly reflect the regional tectonic regime present in the island (see Fig. 2), while the above-mentioned volcano–tectonic lineaments may denote a more local tectonic control associated with the São Sebastião crater genesis, as expressed by its radial pattern.

SEISMICITY IN THE SÃO SEBASTIÃO AREA

Besides a more or less continuous microearthquake activity recorded on the regional and the local seismographic arrays, Terceira Island has been affected by several moderate to strong earthquakes, such as the Praia da Vitória earthquakes in 1614 and in 1841 and the 1980 January 1 earthquake, the strongest in the Azores during the last century, with $M = 7.2$ (Hirn *et al.* 1980), which caused significant damage, injuries and severe economic disruption (Oliveira 1992). Additionally, this island is often affected by seismic crisis or swarms with dozens or hundreds of events, as was the case for the 1997 June–October swarm, with a mainshock with magnitude $M_b = 5.6$ that hit mostly the SE area of Terceira with maximum intensities of V on the Mercalli modified intensity scale. The swarm epicentres were located on the western flanks of the D. João de Castro Bank, which is placed between the S. Miguel and Terceira islands (Oliveira *et al.* 1997).

Most of these and other earthquakes were felt in the village of São Sebastião with higher intensities (MMI) than in the neighbouring communities located outside the depression, such as Ribeira Seca and Porto Judeu (Fig. 3) (e.g. Forjaz 1980a; Oliveira 1992). Moreover, instrumental data clearly show that the earthquakes recorded inside the depression have higher peak ground accelerations than those recorded elsewhere in the vicinity of the depression. For example, the 1998 July 9 earthquake (with epicentre near Faial island and $M_b = 5.8$), was recorded at the São Sebastião strong motion station with peak ground acceleration values twice of those recorded at Praia da Vitória (both located almost 125 km from the epicentre) and slightly higher than those measured at Angra do Heroísmo, 11 km closer to the epicentre (Oliveira *et al.* 1998).

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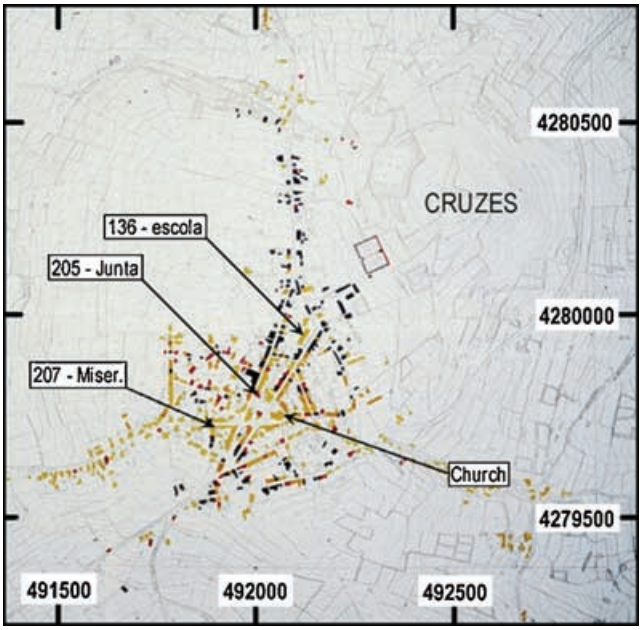

Figure 5. Geographical distribution of damage in São Sebastião village for the 1980 January 1 earthquake. The inset arrows show the location of three strong motions stations inside the depression (136-'Escola', 205-'Junta de Freguesia' and 207-'Misericórdia'). UTM coordinates are in metres.

This anomalous seismic behaviour also has a more local expression, inside the depression itself, shown for example during the strong 1980 January 1 earthquake, when the heavily damaged buildings were located mostly in the NE sector of the depression, rather than in the central part of the village close to the church (Fig. 5). These circumstances were also confirmed by several measurements made with strong motion stations located at three different sites inside the crater (Escola, Junta de Freguesia and Misericórdia, Fig. 5), 200 m apart. In several events associated with the seismic crisis of 1997 June–October (Oliveira *et al.* 1997), the amplitude recorded at Escola (Site 136) reached values twice those recorded at the other two sites, Junta de Freguesia (Site 205) and Misericórdia (Site 207). Fig. 6 is an example of this site effect for a single event, clearly showing that the three records (N–S component) have quite different signatures with the amplification at Escola lasting for a longer period of time.

In general terms, some of the local geological conditions, such as loose and soft material filling the São Sebastião crater, contrasting with the hard lava flows in its borders and vicinity (Fig. 4), could be called on to explain the above-mentioned regional anomalous seismic behaviour. It also seems reasonable to consider that the amplification of the seismic waves inside the crater is, to some extent, related to the general shape and configuration of this basin-like structure, to the softness of the filling material and to the wave reflection process that takes place. Such a phenomenon on a larger scale is widely documented in various cases (Lomnitz 1988; Aki 1993; Makra *et al.* 2001; Safak 2001).

GRAVITY AND TOPOGRAPHIC DATA: REDUCTION AND ANALYSIS

The gravity data consist of 412 observations (according to close itinerary and redundant measurements to control instrumental drifts, jumps, etc.) made during two survey campaigns in 1998 and 1999,

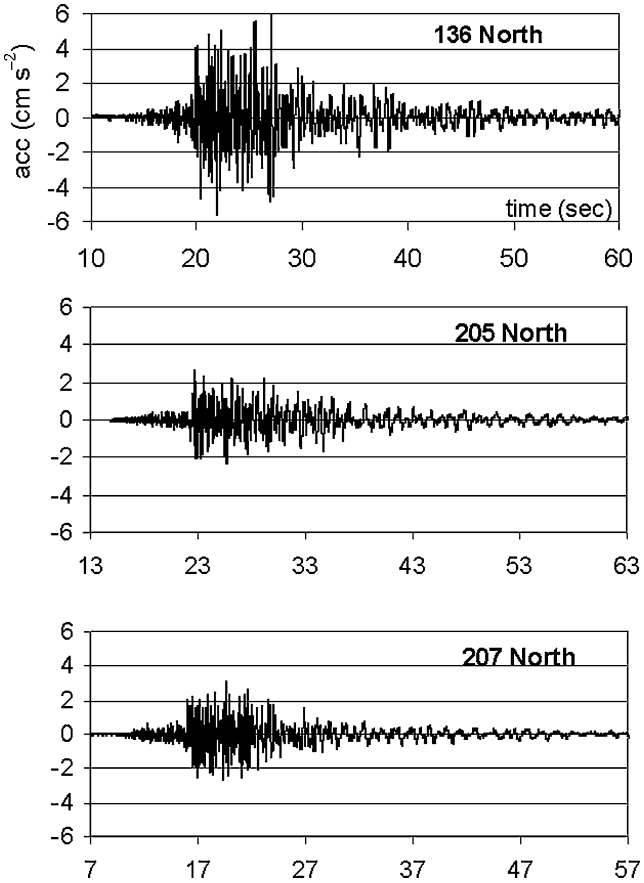

Figure 6. Time history recorded at three strong motions stations (N–S component, 1998 July 15, 21:00:43, $M_D = 4.1$) during the 1997 seismic crisis in three sites of São Sebastião: (136) Escola; (205) Junta de Freguesia; (207) Misericórdia (see Fig. 5).

at a total of 334 sites, using a G-933 LaCoste–Romberg gravimeter with digital electronic readings. The distance between stations is about 200 m for the regional study and about 50 m in the urban zone inside the crater (Fig. 7).

The gravity data were first corrected taking into account the tidal model calculated with the records of the tidal gravity stations on São Miguel Island (Vieira *et al.* 1996). Afterwards, using the redundant observations, a global least-squares adjustment was made, taking linear drifts, possible record jumps (as a result of mechanical or thermal shocks), and the (incremental) gravity values for each station as unknowns. This adjustment led to residues (observational uncertainties) with a standard deviation of 0.021 mGal (1 mGal = 10^{-5} m s^{-2}). Furthermore, the adjusted gravity data were referenced to the value 980 176.860 mGal, as determined by the Portuguese Geographic Institute (Coelho 1968) at the station Cais da Figueirinha on Terceira Island.

The 3-D station coordinates were obtained using trigonometric calculus from topographic survey observations (planimetry and altimetry) performed from geodetic marks. The accuracy of these determinations is about 0.01 m.

To include the topographic correction in the gravity data, a digital terrain model (DTM) was developed for the zone of São Sebastião and surrounding areas, including bathymetric data, by means of a dense regular digitizing of 1:2000 topographic maps; the effect of more distant relief was determined from 1:50 000 maps. Fig. 7 shows an orthographic view of the resulting terrain model (81 829

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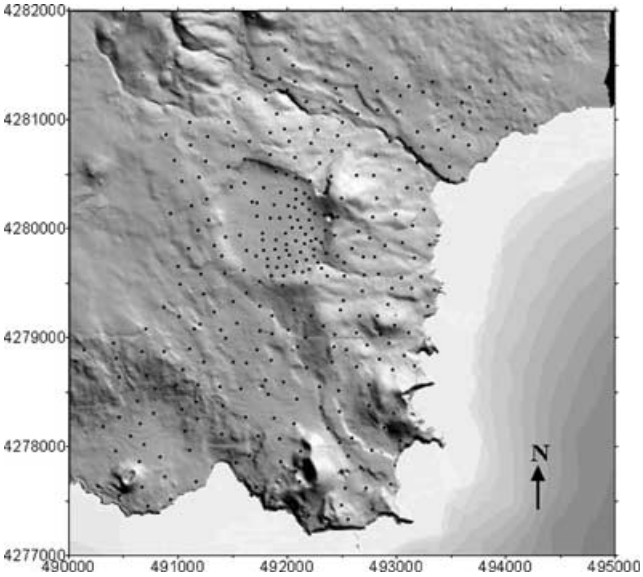

Figure 7. Location of the 334 gravimetric observation sites (dots), stated over a digital terrain model of the studied area. The highest point corresponds to 335 m. UTM coordinates are in metres.

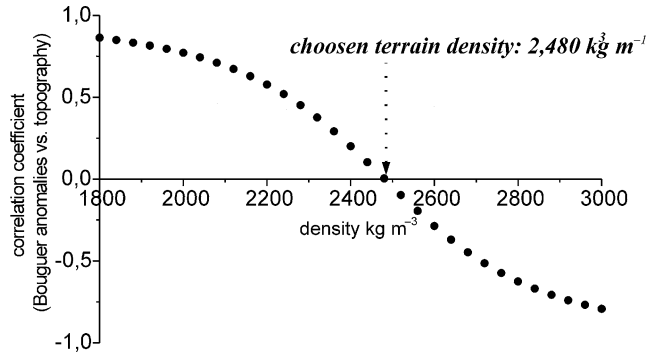

Figure 8. Selection of the terrain density value 2480 kg m^{-3}, according to a similar criterion to the Nettleton's method, corresponding to the minimum correlation between the Bouguer anomalies and the topography.

data points) of the survey zone, which reaches a maximum altitude of 335 m. In this DTM, the São Sebastião crater rim is clearly identified as well as several scoria cones (and their craters), including those located near the São Sebastião crater borders. The Porto Novo fault, with a general NW–SE trend (see Fig. 4), and some of the major streams in the studied area (top left of Fig. 7) also have a clear morphological expression in this model.

The Bouguer gravity anomaly map was obtained using the 1980 normal gravity formula, and the terrain effect was corrected out to 20 km, according to the influence of the relief versus its distance from gravity stations. A density value of 2480 kg m^{-3} was calculated by means of a similar criterion to that described by Nettleton (1939) (e.g. Montesinos *et al.* 1999). Nettleton estimates the terrain density by making a special traverse of gravimeter stations across a topographic feature, reducing these stations for several different densities and finding the density value for which the Bouguer anomaly curve has a minimum correlation with the topography, assuming that there is no associated subsurface geological structure that might cause a gravity anomaly coextensive with the topographic feature. However, in a 3-D study it is difficult to select a representative profile for the whole area. Moreover, in a volcanic zone the previous hypothesis concerning a null correlation between topography and the geological feature is improbable. For these reasons, instead of using only a profile we achieve a global analysis of the whole data. Though this criterion is not strict either, in the sense that there are several topographical masses that might be associated with sources of the gravity anomaly field, the terrain density obtained can be a reasonable guide value as an average density of all the material within the elevation range of the gravity stations. Finally, this density value must be contrasted with other geological information to be adopted.

We have computed Bouguer anomaly maps using a set of terrain densities ranging from 1800 to 3000 kg m^{-3}, looking for minor associations with the topography. By means of calculating the correlation coefficient (or Pearson's correlation coefficient) (Fig. 8), the minimum correlation corresponds clearly to 2480 kg m^{-3}, which is only a possible average density of all the material within the elevation range of the gravity stations. However, this minimum correlation value agrees with those proposed by several authors. It is slightly smaller than the density of 2600 kg m^{-3} chosen by Ridley *et al.* (1974) in their preliminary gravity study of Terceira Island, but it nearly corresponds to the density of 2500 kg m^{-3} assumed by Self (1976) for the silicic rocks (comenditic trachytes to pantellerites) that cover the area north of the São Sebastião crater (Fig. 4). Moreover, the density value of 2480 kg m^{-3} can be considered as a mean value between those proposed by Hyndman *et al.* (1979) for the volcanic islands. In fact, these authors conclude that mean densities as low as 2300 kg m^{-3} and as high as 2600 kg m^{-3} characterize the subaerial volcanic products that make up the upper part of most active volcanic islands.

Experimental density determinations carried out for the volcanic rocks of Azores indicate the values 2250 to 2840 kg m^{-3} for lava flows of a basaltic nature (either being vesiculated or massive, respectively), while for the corresponding products of an explosive nature (e.g. basaltic scoria deposits) these values drop to about 1300 kg m^{-3}. Additionally, the more evolved (silicic) rocks, such as trachytic lava flows, show average density values of about 1800 kg m^{-3}, with a maximum value of 2425 kg m^{-3} found for a lava flow of comenditic trachyte. This value agrees well with that proposed by Self (1976) for Terceira Island. Although not dominant in this area of Terceira, it should also be mentioned that pumice deposits (due to its high degree of vesiculation) gave standard density values of about 600 kg m^{-3}. A critical review of all of these determinations within the framework of the present work must emphasize that for the subsurface levels in the São Sebastião area (several tens or hundreds of metres below sea level) the geological layers should be composed primarily of more or less thick lava flows, of a basaltic and trachytic nature, with a significantly smaller percentage of explosive products (scoria and pumice). So, according to these determinations, the value of 2480 kg m^{-3}, obtained by this version of the method of Nettleton, can be considered near to the average value of the density of the materials in the area.

Then, taking a terrain density value of 2480 kg m^{-3} and a water density of 1030 kg m^{-3} for the terrain correction for the bathymetry, the evaluated topographic correction ranges between 1.0 and 4.3 mGal, and the resulting Bouguer anomaly Δg (Fig. 9) varies between 136.5 and 141 mGal. On the Bouguer anomaly map an alignment of gravity minima, which begins inside the crater and continues to the south coast, with a general NNW–SSE trend, is clearly observed. A main gravity maximum zone with a NW–SE trend appears NE of the studied area, outside of the crater, while another gravity maximum is located near the northern border of the

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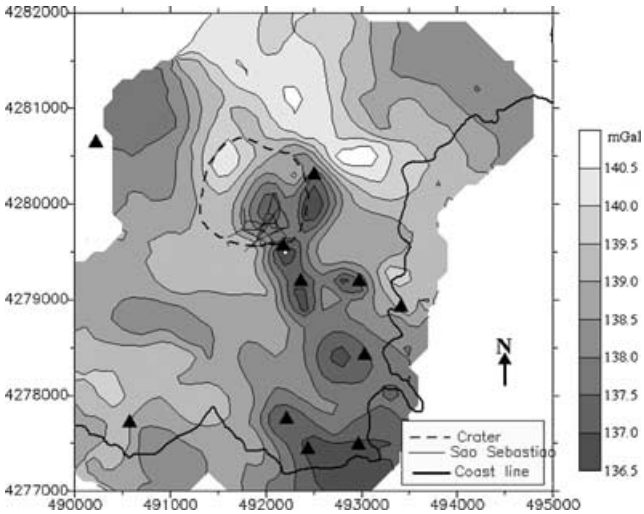

Figure 9. Bouguer anomalies map corresponding to a terrain density value of 2480 kg m^{-3}. Triangles point out scoria cones eruptive centres (see Table 1 and Figs 4 and 7). UTM coordinates are in metres.

crater. Finally, a weaker high-gravity zone runs along the western half of the São Sebastião crater.

To estimate the noise in the gravity values corresponding to observational uncertainties and to inaccuracy in the data corrections (namely topographic correction), we applied a covariance analysis of the gravity data and a further least-squares prediction (or filtered of the signal) (e.g. Montesinos *et al.* 1999; Moritz 1989). This process takes into account the fact that the relation between the observed data and an estimation of the signal is given in terms of the covariance and the noise and the signal are uncorrelated. By means of a correlation analysis of the data and assuming that the covariance between two stations depends only on the horizontal distance, we calculate a distribution of empirical covariances, which is adjusted by means of a suitable analytical covariance function (Barzaghi & Sanso 1983). The usual formulae for least-squares prediction (Moritz 1989) give us a first estimation of the signal at this level of covariance and the corresponding prediction error matrix. By considering the residual values (observed data minus estimated signal), the non-correlatable part for the previous covariance level, we again repeat this process with a new covariance analysis in order to detect, if possible, correlated signals of minor amplitude. This process ends when no further signal can be detected. Finally, by eliminating these signals detected from observed data, the distribution of resulting residual values displays the non-correlatable noise. In this case, we estimate the signals (three levels) corresponding to the gravity anomaly field (such as a correlated part of the data) with a standard deviation of 0.213 mGal for the non-correlated noise (Fig. 10). This noise may be related to observational noise, very local anomalies or uncertainties in the previous corrections (such as terrain corrections) and its value validates the good quality of the data, as required for a study of this type. Moreover, from the error matrix of the estimated signal by means of this process, we calculate a covariance matrix, $\mathbf{Q}_D$, for the data, which is used in the inversion process that follows.

GRAVITY INVERSION

Inversion of gravity anomalies consists in using the observed data to generate a model which reproduces, as well as possible, subsurface geological structures that have a density contrast with their surroundings. Among the several possibilities, we determine the ge-

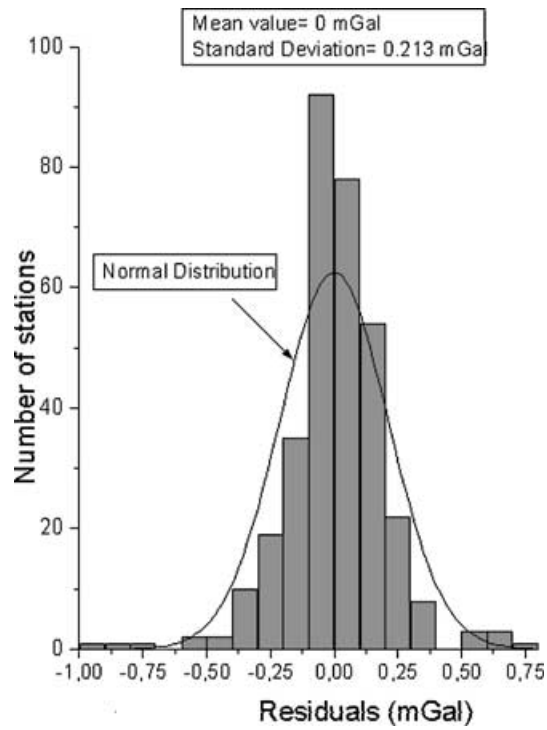

Figure 10. Histogram of the residues of the Bouguer anomalies analysis. These residues correspond to noise detected in the observed data and fit a normal distribution of mean value 0 mGal and a standard deviation of 0.213 mGal.

ometrical properties of these anomalous bodies by fixing the density contrasts. There are different methods, proposed by several authors, for solving this non-linear inverse problem. For example, the traditional local optimization techniques start from initial models and improve them iteratively (e.g. Enmark 1981; Barbosa *et al.* 1997). The main problem occurs when the initial model is not appropriate, because the final solution will depend on this initial information. Moreover, the problem becomes more complex for 3-D models and when positive and negative anomalous contrasts are simultaneously allowed. To avoid these problems, in the case of a few model parameters, one possibility consists in searching for the best solution through a predefined space model (Tarantola & Valette 1982; Walter 1993). Nevertheless, when the number of parameters does not allow the application of this global search, it is necessary that the exploration process be conducted in some other way, either randomly or according to some search rule (Silva & Hohmann 1983; Montesinos 1999). In this study, we use an inversion method, described in Camacho *et al.* (2000, 2002), belonging to the latter group.

We consider the subsurface anomalous structures formed by masses with prescribed positive or negative density contrasts. The problem consists in dividing the whole subsurface volume into a discrete 3-D partition of m prismatic elements to determine the geometry of the anomalous bodies corresponding to those density contrasts. The size and resolution of this partition are related to the diameter and to the step of gravity the stations distribution. Then, the anomalous volumes are constructed by means of an 'expansion approach' testing, step by step, the prismatic elements with all prescribed densities, and adding the best options to the growth approach. The choice of the density contrast values that correspond to sources of the gravity field is not a very critical question. If we adopt a smaller density contrast the resulting model would be rather

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similar, but with more inflated volumes perhaps unable (taking into account the volume limitations) to fit the main components of the gravity field. This result contrasted with the geological information to help remove inadequate density contrasts. Thus, a suitable choice of the *a priori* density contrasts requires the testing of several values looking for compact structures according to the available geological information and the best fit to the observed gravity data.

We begin our inversion process assuming that there is no previous information concerning the model, so we choose as a starting model all cells with null density contrast. If there is some information available it is possible to introduce it on the initial model, fixing a distribution of density contrasts to several prisms, but without the possibility of changing.

After a step-by-step approach, the final structural model will be obtained by assigning to each jth prism, $j = 1, \ldots, m$, a density contrast $\Delta\rho_j$. Nevertheless, these adjusted anomalous density contrasts of the prisms, which are modified, can have only two previously prescribed values: positive or negative. For the 'unchanged' prisms we assume a null value for the corresponding $\Delta\rho_j$.

This inversion technique is based on a global optimization method by means of searching the global minimum of a misfit function.

Let us consider n gravity stations located at $P_i\ (x_i, y_i, z_i)$, $i = 1, \ldots, n$, with observed anomalous gravity values g_i. The gravity attraction, A_{ij}, at the ith station, due to the jth prism, for unit density, is according to Pick *et al.* (1973):

$$\begin{aligned} A_{ij} = -G\Big[\Big[\Big[&x \ln\left(y + (x^2 + y^2 + z^2)^{1/2}\right) \\ &+ y \ln\left(x + (x^2 + y^2 + z^2)^{1/2}\right) \\ &+ z \arctan\left(z\left(x^2 + y^2 + z^2\right)^{1/2} x^{-1} y^{-1}\right)\Big]_{u_1^j - x_i}^{u_2^j - x_i}\Big]_{v_1^j - y_i}^{v_2^j - y_i}\Big]_{w_1^j - z_i}^{w_2^j - z_i}, \end{aligned} \tag{1}$$

where G is the gravitation constant, the edges of the jth prism are parallel to the reference axes and the limiting coordinates for its volume are: u_1^j, u_2^j for the x coordinate, v_1^j, v_2^j for the y coordinate and w_1^j, w_2^j for z. The matrix, $\mathbf{A}$, with components A_{ij} is the design matrix of the direct problem.

In each step of the inversion process, the method modifies the densities of the prisms so that the modelled gravity anomalies fit, with a scalefactor f, the observed gravity anomalies. In other inversion schemes, the presence of a regional component of the gravity data that masks the effect of local structures can perturb the interpretation of the gravity data. However, in this inversion process it is also possible to fit a regional trend of the gravity data by means of a polynomial surface.

So, for an arbitrary $(k + 1)$th step, $k = \ell_1, \ldots, \ell_k$ prisms have been definitively modified, with the positive or negative fixed values. Now, we look for one new jth prism, $j \neq \ell_1, \ldots, \ell_k$, to modify throughout the $m - k$ unchanged prisms. To test the positive and the negative prescribed density contrast for this prism, we consider, first, one of these values, $\Delta\rho_j$, and then we minimize the corresponding residuals given by

$$\begin{aligned} g_i - \left(\sum_{k=\ell_1}^{\ell k} A_{ij}\Delta\rho_k + A_{ij}\Delta\rho_j\right) f - (p_0 + p_x(x_i - x_M) \\ + p_y(y_i - y_M)) = v_i \qquad i = 1, \ldots, n, \end{aligned} \tag{2}$$

where x_i and y_i are the planar coordinates of the ith station, x_M and y_M are the coordinates of an arbitrary central point for the survey (for example, mean coordinates), f is an adjustable scalefactor that keeps the model gravity nearly proportional to the observed values, g_i, and p_0, p_x and p_y are three unknown values which adjust a simple regional trend.

To avoid the characteristic non-uniqueness problem induced by the acceptance of positive and negative values and the inclusion of the unknown regional trend in the model, we do not use a single minimization criterion for the residuals. The solution is obtained by minimizing the misfit function from a mixed condition formed by the gravity residuals (l_2-fitness) and the whole identified anomalous mass quantity, using a parameter λ (positive) for the suitable balance, and fixed empirically (Camacho *et al.* 2000):

$$\mathbf{v}^{\mathrm{T}}\mathbf{Q}_{\mathrm{D}}^{-1}\mathbf{v} + \lambda f^2 \mathbf{m}^{\mathrm{T}}\mathbf{Q}_{\mathrm{M}}^{-1}\mathbf{m} = \min, \tag{3}$$

where $\mathbf{v} = (v_1, \ldots, v_n)^{\mathrm{T}}$ is the vector of the gravity residuals for the n stations; $\mathbf{m} = (\Delta\rho_{\ell_1}, \ldots, \Delta\rho_{\ell_k}, \Delta\rho_j)^{\mathrm{T}}$ is the model parameters vector corresponding to the density contrasts for the prisms that have been modified in previous steps and the density contrast for the prism being tested; f is the unknown scalefactor; and $\mathbf{Q}_D$, $\mathbf{Q}_M$ are suitable covariance matrices corresponding to the estimated inaccuracies for the gravity data and for the determinability or sensibility of the 3-D grid cells from the gravity stations. The matrix $\mathbf{Q}_D$ is determined in the previous covariance analysis of the gravity data. If there is no previous information concerning the model structure, we suggest (Camacho *et al.* 1997) taking a model of covariance matrix $\mathbf{Q}_M$ given by a diagonal normalizing matrix of non-null elements that are the same as the diagonal elements of $\mathbf{A}^{\mathrm{T}}\mathbf{Q}_D^{-1}\mathbf{A}$. So, the matrix $\mathbf{Q}_M$ is defined as a function of the gravity attraction at the stations, due to the prisms and, therefore, of the distance between the points and the prisms and of the volume of these prisms. The elements of this matrix can also be used to limit the whole subsurface volume under study, bounding the accepted model uncertainty. Taking into account the fact that the covariance matrix $\mathbf{Q}_{\mathrm{M}}^{-1}$ contains the prism volumes as factors, the second term in the minimization condition (3) is connected with the anomalous mass of the model, giving rise to models which, for a level of fitness, look for compact volumes.

The solutions of the equation system (2) and (3) can be calculated for the adopted prism and $\Delta\rho_j$ as is indicated in Camacho *et al.* (2000). The same process is repeated by choosing the other possibility of the density contrast value. Finally, the jth prism and tested density (positive and negative) which produce a minimum value of the misfit function (3) are definitively selected to increase the model but only if this value is less than the former step.

The parameter λ controls the application of the minimization conditions and is chosen so that the standard deviation of the inversion residuals is similar to the data noise level (0.213 mGal) determined by the previous covariance analysis.

So, from the starting model with all prisms with null density contrasts, this process is repeated successively. For the consecutive steps, the scale value f decreases and the trend parameters p_0, p_x, p_y, reach nearly stable values. The process stops when f approaches 1, resulting in some distribution of the prescribed density contrasts representing the sources of the gravity anomaly field and a final regional trend.

The solution obtained in this inversion method, detailed in Camacho *et al.* (2000, 2002), will depend on a suitable choice of the *a priori* density contrasts (positive and negative). Higher density values will produce an anomalous model with small structures and more compact than if lower contrast values are chosen. Logically, the adjusted geometry for these anomalous bodies can appear too sharp, being controlled by the selected density contrasts. To produce a model with a smoother geometry that can diminish the uncertainty of the inversion problem, we employ a smoothing technique. This appears in the final model structures with a small gradient of density contrasts that blurs the borders of the anomalous bodies.

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Also, we must take into account that the gravity inversion is insensitive to the effects of horizontal stratification. So, the solution obtained by means of this inversion method, considered without any possible stratification, offers interesting information concerning the geometry and location of the anomalous sources of the gravity field.

RESULTS AND DISCUSSION

We applied the gravity inversion method in the area of São Sebastião to determine the geometry of the shallow crustal structures with density contrast with respect to the surroundings. Taking into account the distribution of gravity stations and the distance among them, we have calculated two 3-D models that help us to identify deeper or regional structures and shallower or local bodies, respectively. While the regional survey covers a region of 17.5×10^6 m^2 surrounding the volcanic depression and considers all the gravity stations, the local study was carried out at a surface of 3.4×10^6 m^2 with 252 gravity stations. So, for the first model we assumed 17 469 prismatic elements (cubes) with an average side of 200 m, making a grid of the subsurface volume down to a depth of 1500 m. Nevertheless, if the objective of this work were to establish fully deeper sources, more extensive surveys would be needed. To better resolve local structures, the second model consists in 16 408 prismatic elements, with an average side of 50 m (in this zone the distance between contiguous stations is about this value) and reaching a depth of 400 m.

To decide the values of density contrasts for the structures that we want to identify by means of the inversion technique, we tested several values looking for the best correlation between the compactness of structures and the geological information and the best fit to the observed gravity data. We selected -500 and $+350$ kg m^{-3} as extreme density contrasts, which should correspond to usual contrasts among volcanic rocks in the São Sebastião depression area. These values seems to be compatible with the main density contrasts expected in the area considered, as mentioned in previous chapters, maximum density values of 2840 kg m^{-3} for the basaltic lava flows and of 2425 kg m^{-3} for those with thrachytic composition, which represent the formations with more important subsuperficial expression. Nevertheless, using different density contrasts, the solutions are rather similar although for higher contrast, smaller volumes are obtained.

The structural models determined from gravity inversion are represented by several horizontal sections at different depths and by a few vertical profiles (Figs 11 and 12) at convenient locations. The distribution of density contrasts is shown in horizontal sections (referred to sea level, downward z negative) with smoothing of the final model obtained in the inversion process.

The gravity residuals from inversion for the regional model fit a normal distribution with a mean value of zero and a standard deviation of 0.255 mGal. This value is similar to the standard deviation of the data noise level determined from the covariance analysis for all the data (0.213 mGal). The residuals from the inversion for the local model have a smaller standard deviation of 0.025 mGal. This is due to a better distribution of stations, with a smaller distance between them and a smaller range of corresponding heights.

With respect to the regional model (Fig. 11), several lineaments and anomalous bodies (either with positive or negative contrasts) can be identified and some of which may be correlated to the surface geology. According to the characteristics of the gravity survey, the anomalous sources of the gravity field are detected until depths of about 700 m. In deeper sections a few bodies can be only lightly suspected. Between the -100 and -300 m horizontal sections the positive contrast structures form two main alignments, shown in Fig. 11 as 'a' and 'b' (section $z = -200$ m). The first one, with a general NW–SE trend, runs along the surface trace of the Porto Novo normal fault (see also Fig. 4). This alignment is composed of two similar main bodies located on the SW collapsed block of the fault, and are observed in all horizontal sections of the model, until depths of about 600 m. On the NE block of the fault, the model identifies a body with low density, which is shallower and smaller than the surrounding structures and that contrasts with the above-mentioned high-density bodies, which points out the NW–SE fault trend.

The second alignment of high-density bodies ('b' in Fig. 11) is located to the south of the crater, starting at its southwestern wall and extending until the southern coast, along Canada da Ponta area (Fig. 4). These bodies with positive density contrast do not have a clear morphological expression at the surface and they do not appear in the deepest sections of the model, except for the body located near the SW rim of the crater, which reaches depths of about 400 m. Taking into account the volcanic framework of the area, the former shallow dense bodies might correspond to subsurface intrusions associated with 'intermediate-age' flows, of fissural nature, extruded in the area (Fig. 4). On the other hand, the above-mentioned SW crater border positive anomaly, together with a positive density contrast body identified near the NW margin of the São Sebastião depression (Fig. 11), may correspond to the bordering of the structures affected by the collapse that formed the crater. It can be observed that this characteristic is not found at the eastern part of the crater, because the gravimetric signature is mostly expressed by low-density formations, namely scoria cones, dominant in the area (Fig. 4).

There is a unique body with a positive density contrast in the model that can be correlated with a volcanic edifice of the area: the remains of the scoria cone 'E Arrabalde' ('6' in Fig. 11). In spite of the peripheral position in the model of that body and, therefore, its careful interpretation, the presence of the 'E Arrabalde' pyroclastic cone near the seashore, deeply eroded and with an exposed neck of dense rocks, might explain this positive density contrast of the model.

An alignment with a general NW–SE trend of negative density contrast ('c' in Fig. 11) is identified in the inversion model sections, and corresponds to a volcanic alignment of scoria cones that extends from the southern border of the crater to Pico dos Comos ('9', '8' until '5' in Fig. 11; see also Fig. 4). Most of the individual negative anomalous bodies revealed by the 3-D model, which reach depths of about 300–400 m, identify the location of these and other scoria cones ('2' to '10', in Fig. 11 and Table 1). These low densities are due to the deposits associated with these cones, which are built-in by highly vesiculated material of basaltic nature, associated with moderate explosive activity of strombolian type, while the surroundings are basaltic lava flows with higher densities.

The body with a negative density contrast located inside the crater ('d' in Fig. 11) seems to be the most important deep structure with low density in the model. This body is detected in the horizontal sections until depths of about 600 m and is slightly displaced to the SE of the depression. Its circular shape and its position indicate that this source is directly related to the genesis mechanism of the crater (e.g. Rymer & Brown 1986). Given the low density of the loose and detritical deposits that infill the São Sebastião depression (e.g. unconsolidated ashes, pumice, scoria, slope and alluvial deposits, to which densities of about 600–1100 kg cm^{-3} are expected) the presence of this low-density body might be correlated with the

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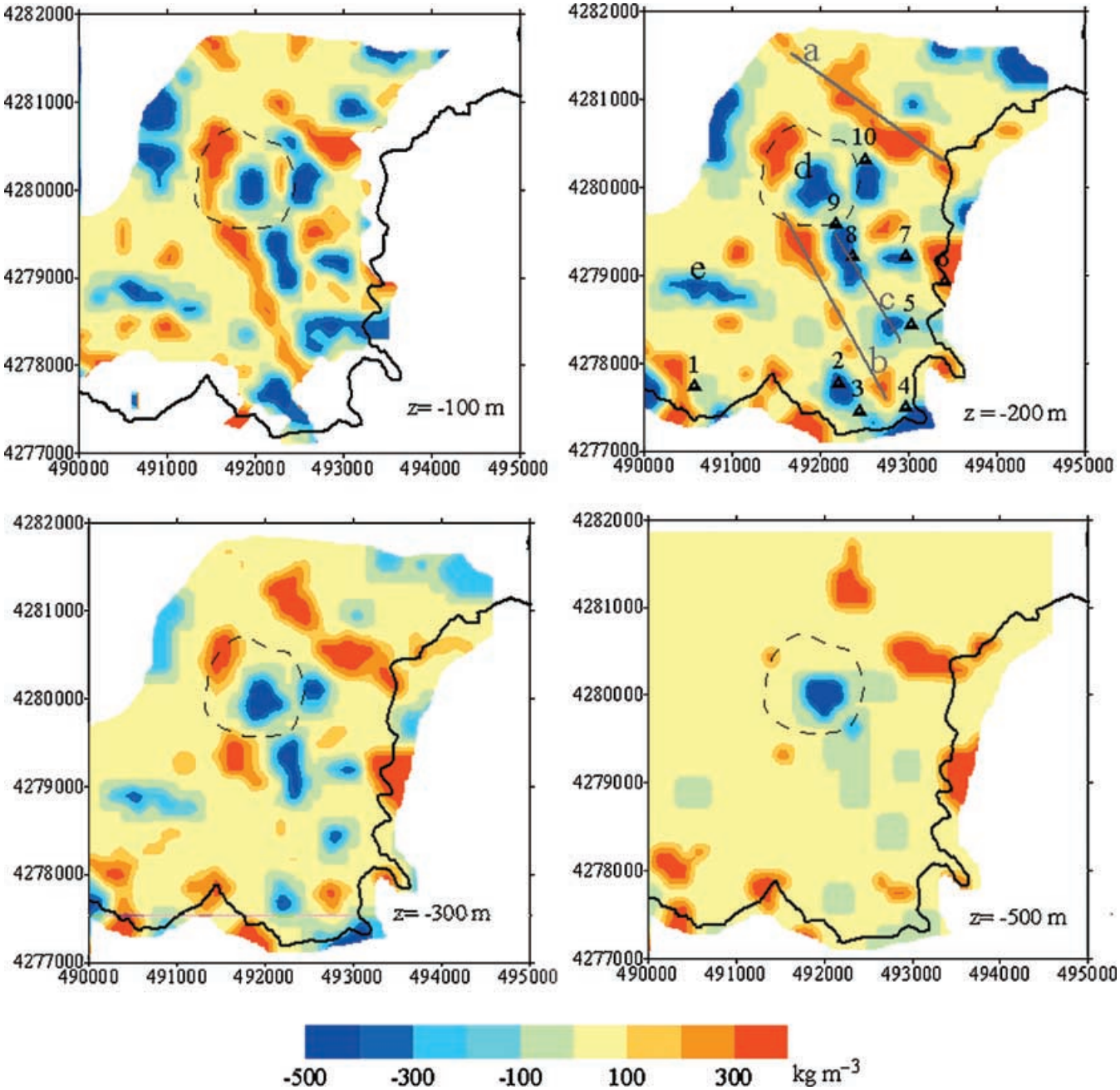

Figure 11. Deep horizontal sections of the density contrast 3-D regional model (for −100, −200 and −500 m) for the whole studied area. The broken line indicates the crater rim and the solid line points out the coastline. The section for −200 m includes a general interpretation about the major volcanic and tectonic lineaments deduced for the area. In this section, triangles point out scoria cones eruptive centres and solid thick lines proposed lineaments. For reference numbers and letters see the text and also Table 1. UTM coordinates are in metres.

post-eruptive collapse and the later infilling of the crater with that loose material (Fig. 4). Although it is obvious that to better define the deep configuration and nature of that body a more extensive survey would be needed, the geological and gravity data now presented clearly indicate a shallow mass deficiency in this area, associated with the uncompacted crater infill.

Additionally, the gravity inversion model shows a significant low-density body located towards the southwestern part of the studied area ('e' in Fig. 11), with a general E–W trend, and that reaches depths of about 300 m. This body cannot be directly correlated with the surface geology and tectonics of the area. However, its location is close to the shoreline on the area before the Pico do Refugo eruption, which formed that scoria cone and its lava delta (Fig. 4). The primitive shoreline can be identified on the DTM model of Fig. 7, as the area with a higher slope north of that cone. On the other hand, the clear W–E trend of that low-density source might suggest the presence of a fracture in the area, with no superficial expression.

The interpretation of all the peripheral bodies (positive or negative) of the regional density contrasts model must be made with caution because of their proximity to the boundaries of the surveyed area. This is true for the low-density body shown at the NW of the

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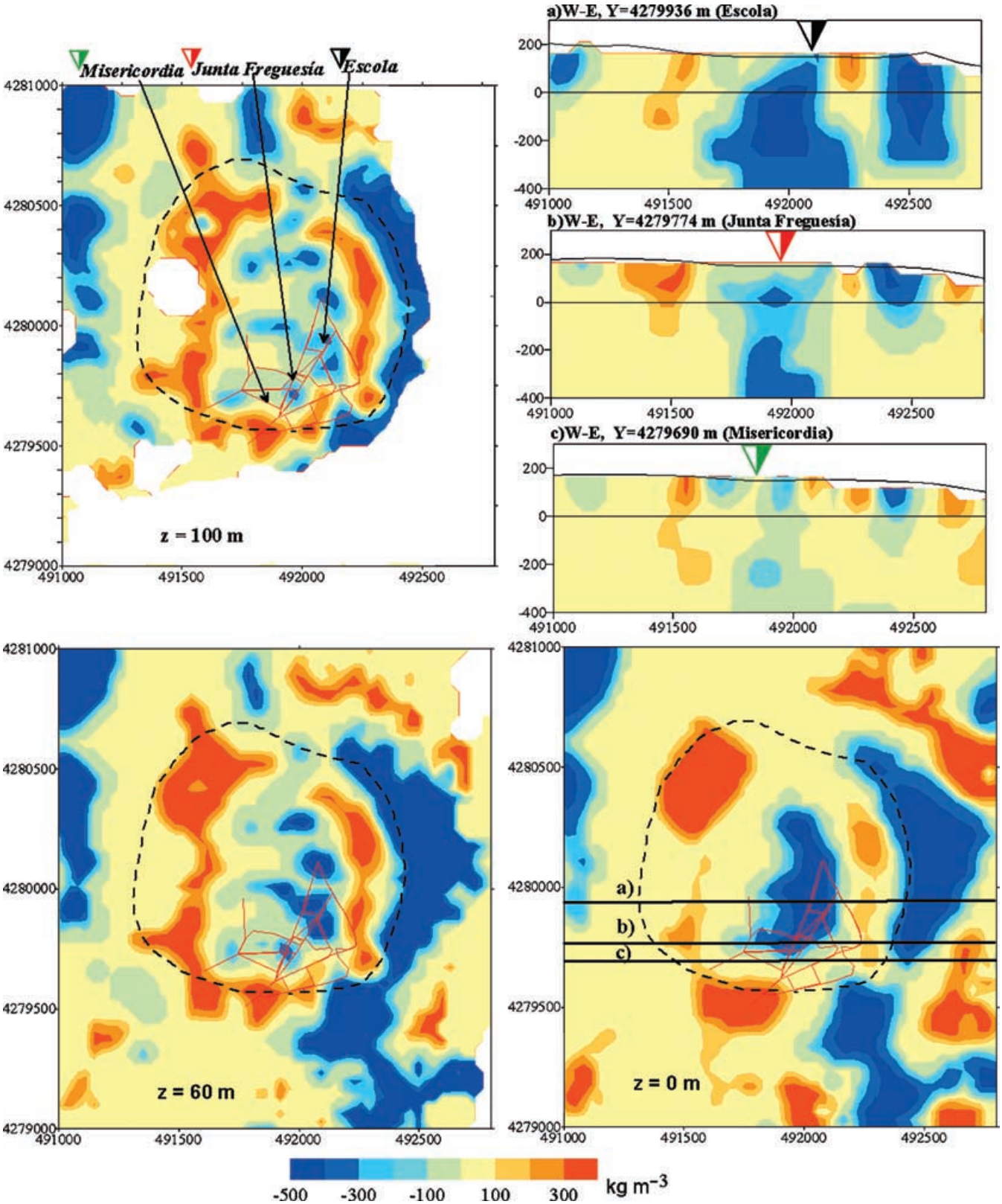

Figure 12. Horizontal section of the density contrasts 3-D local model (for 100, 60 and 0 m a.s.l.) and E–W profiles along the lines shown in the $z = 0$ m horizontal section crossing the locations of the three stations with seismic records. In the section $z = 100$ m and in the vertical profiles the location of these seismic stations is indicated. Inside crater, red lines points out the main routes in the urban centre of São Sebastião. The white areas are the effect of a greater separation between stations that produces a lack of information at that level. UTM coordinates are in metres.

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gravity model and located outside the São Sebastião depression, as well as for the two significant low-density bodies located in the Fonte Bastardo area (Figs 4 and 11). The later might reflect the youthfulness of the volcanism of that area (2000–3000 years old).

More detailed field work done in the São Sebastião volcanic crater allowed a more precise gravimetric modelling, shown in Fig. 12, that points out several local structures located inside the depression and its vicinity. In the shallower section (100 m a.s.l.), a ring of structures with positive density contrast borders the crater wall, but this feature is not very well constrained on its western side due to the lack of measurements at that border. These bodies are located inside and around the São Sebastião depression, until depths near to sea level. They correspond to a more detailed and shallower expression of the two main high-density bodies referred in the regional model (Fig. 11), which, as mentioned, are associated with the presence of the basaltic lava flows that form the crater wall (Fig. 4) and define the collapsed area.

The spatial and size relationships between these dense bodies and the present depression size can be associated with the original location of the crater walls: these, formed by basaltic lava flows, were later affected by important erosional processes (including rockfalls and other mass movements), which widened the crater rim and produced the important slope deposits that presently border the depression. Once the geometry, the size and the location of these high-density bodies is rather similar at all the horizontal sections of the model, it seems reasonable to conclude that the morphology of the São Sebastião depression is characterized by very steep walls, in agreement with the collapse mechanism proposed for the genesis of the crater (Nunes 2000).

As already pointed out, the bodies with negative density contrast present in the inner area of the crater correspond to the infilling and very soft material accumulated inside the depression after its formation. However, the collapse itself also justifies the existence of that low-density body inside a dense structure ring (slightly displaced), as well as its deep signature (Figs 11 and 12). In fact, these collapses and the inherent instability of the volcanic surface usually gave rise to the formation of breccia-type deposits on the conduit system, which are identified as negative density contrast bodies.

In the shallower sections of this more detailed model, several E–W and N–S trends are clearly seen for those low-density bodies located inside the crater (Fig. 12), forming an inverted 'E'. Once these alignments (not due to the gravimetric stations distribution, or observations sequence, etc.) cannot be correlated with any surface geological feature, it seems reasonable to admit that the 'E' pattern mentioned may reflect the existence of a fracture system inside the depression, possibly generated by the collapse of the crater. It should be stressed that the thick sequence (about 38 m) of detritical and loose deposits superimposed on this probable fracture system, generates zones with a mass deficiency in relation to the surrounding denser formations, clearly identified on the shallower sections of this gravity inversion model of São Sebastião depression.

In order to better understand the anomalous seismic response in the São Sebastião area, we looked for possible correlations between these low-density structures and the different site effects produced by seismic events in the village itself, and in the São Sebastião crater with respect to the neighbouring communities, such as Ribeira Seca and Porto Judeu.

The local 3-D gravity inversion model (Fig. 12) confirms that the São Sebastião village subsurface is dominated by low-density structures, which are confined inside the crater and flanked by several denser bodies. This settlement generally reflects the local geological conditions (Fig. 4) and its morphogenetic constraints (e.g. a collapsed crater) defining a weak zone in São Sebastião depression. This influences the seismic response observed in the area, with the seismic intensity at São Sebastião systematically reaching a value higher than the neighbouring communities.

According to Hildenbrand *et al.* (1996) such an anomalous behaviour during seismic crisis may be because the structures beneath the crater act as an elongated elastic body with lower rigidity than the surrounding basement. Additionally, the high–low density contrast existing in the area seems to form a 'wall' to seismic waves, thus making the crater a resonance basin (Sincraian & Oliveira 2001).

On the other hand, the anomalous seismic behaviour inside the village (Figs 5 and 6) was analysed with the microgravimetry study, with a spacing of gravity stations of about 50 m, allowing the identification of subsurface structures with more detail. Thus, in the shallowest sections of this local model there is a clear negative density contrast body located beneath the 'Escola site' (Fig. 12) whereas 'Junta de Freguesia' and 'Misericórdia' strong motion sites do not appear to be associated with especially weak or large subsurface bodies and with densities as low as the former. This circumstance is observed in the vertical profiles of the distribution of the density contrast (Fig. 12), where the differences among the subsurface bodies below the three stations are clear. Namely, the low-density body below the 'Escola site' is the biggest and it has connection in surface with the location of the station. In this case, the low-density structure identified in the gravity inversion local model can be associated with the amplified seismic signals recorded at the 'Escola site' (with respect to the other strong motion sites). Given the location, size and trend of those low densities bodies located inside the depression, it also seems reasonable to admit that the unequal distribution of seismic intensities in the village during strong earthquakes (such as during the 1980 January 1 earthquake) (Oliveira 1992) is due to the same circumstance as the São Sebastião crater; i.e. the presence of low-density bodies.

This anomalous behaviour was also shown by an analysis of seismic noise using the Nakamura technique (Senos *et al.* 2000; Pena *et al.* 2001), which detected the presence of low frequencies inside the crater and higher ones outside. These sharp local differences in seismic response correspond in general terms to the great contrast between the very soft material deposited inside the crater and the hard material outside.

As a final synthesis for all the data previously reported, Table 1 summarizes the main volcanic and tectonic structures observed in the area of São Sebastião, as well as the main anomalous bodies and lineaments inferred from the gravity model. It is clear from Table 1 that a good correlation exists, on one hand, between the geological and morphological features and, on the other hand, with the gravimetric results obtained for the area. Additionally, microgravimetric studies performed on Terceira Island allowed us to gain useful information on the definition of deep structures present in the area that sometimes could not be directly identified from the surface and borehole geological data. This might be the case for the tectonic lineament of 'Canada da Ponta' and for the buried scoria cone that borders the crater at the SE (lineament 'b' and '173' cone, respectively, in Table 1).

CONCLUSIONS

This gravity study allows the identification of the main subsurface structures of the volcanic crater of São Sebastião area on Terceira Island. It is focused on the interpretation of the gravity anomalies, relating them with the geology and looking for an explanation

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of the anomalous seismic behaviour in this area. Among the main conclusions given throughout the paper the following points should be emphasized.

(1) The positive contrast structures form two main alignments, the first one runs along the surface trace of the Porto Novo normal fault and the second alignment is along the Canada da Ponta area, which might correspond to subsurface intrusions associated with 'intermediate-age' flows, of fissural nature, extruded in the area.

(2) The high-density bodies located at the SW and near the NW margin of São Sebastião crater, correspond to the bordering of the structures affected by the collapse that formed the crater, and can be associated with the original location of the crater walls.

(3) The morphology of the structures identified in the gravity model is compatible with the collapse mechanism proposed for the genesis of the crater.

(4) There is a good correlation between the structures identified in the gravity study and the existence of volcanic edifices of the area. Almost all of them are associated with low-density bodies except one with an exposed neck of dense rocks. This might explain the high density present in the model.

(5) Inside the crater, the most important deep structure of the model has a low density and it might be directly related to the genesis mechanism of the crater, the post-eruptive collapse and the later infilling. In the shallowest sections, perhaps due to the formation of breccia-type deposits on the conduit system, a distribution of several low-density bodies appears.

(6) The anomalous seismic response observed in the São Sebastião depression regarding the surroundings can be associated with a weak zone dominated by subsurface low-density structures, which are confined inside the crater and flanked by several denser bodies making the crater a resonance basin which produces an important site effect.

(7) Inside the village, the zone which registers an amplified seismic signal with respect to the other strong motion sites is clearly related to the existence of a significant low-density body in the inversion gravity model.

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